

Linear algebra and geometry 1 - TTPU

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State Finished

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Time taken 25 secs

Marks 80.00/1000.00

Grade 2.40 out of 30.00 (8%)

Question 1

Incorrect

Mark -15.00 out of
100.00

In the euclidean space let us consider the points

$$A = (1 + a, 1, 0), \quad B = (2 + a, 1, 0), \quad C = (1, 0, 2), \quad D = (1, 0, 0).$$

Which of the following statements is true?

- ☒ (a) There exists only one $a \in \mathbb{R}$ such that the points are not coplanar. ✗
- ☐ (b) For every $a \in \mathbb{R}$ the points are not coplanar.
- ☐ (c) There exists only one $a \in \mathbb{R}$ such that the points are collinear.
- ☐ (d) For every $a \in \mathbb{R}$ the points are collinear.

Risposta errata.

The correct answer is: For every $a \in \mathbb{R}$ the points are not coplanar.

Question 2

Incorrect

Mark -15.00 out of
100.00Given the plane $\pi : 2x + y - z = 1$ and the straight line

$$r : \begin{cases} 2x + y - z = 1 \\ 3x + 2y + z = 1. \end{cases}, \text{ find the correct statement.}$$

-
- ☒ (a) $r \cap \pi = \emptyset$. ❌
- ☐ (b) $r \subseteq \pi$.
- ☐ (c) None of the other statements is correct.
- ☐ (d) r and π intersect at exactly one point.

Risposta errata.

The correct answer is: $r \subseteq \pi$.**Question 3**

Correct

Mark 100.00 out of
100.00

Let us consider the linear system with 4 variables depending on the real parameter

 $k \in \mathbb{R}$:

$$\begin{cases} x - z = 0 \\ x + 2y - z + 2t = 4 \\ y + t = k. \end{cases}.$$

Find the right statement.

-
- ☒ (a) If $k = 2$ the system admits ∞^2 solutions. ✔️
- ☐ (b) For any $k \in \mathbb{R}$ the system does not admit any solution.
- ☐ (c) If $k = 2$ the system admits ∞^1 solutions.
- ☐ (d) If $k \neq 2$ the system admits only one solutions.

Risposta corretta.

The correct answer is: If $k = 2$ the system admits ∞^2 solutions.

Question 4

Correct

Mark 100.00 out of 100.00

If

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix},$$

then

- ☒ (a) $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$ ✓
- ☐ (b) $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 10 \\ 0 & 0 & -1 \\ 0 & 0 & 1 \end{bmatrix}$
- ☐ (c) $\mathbf{A}^{-1} = \begin{bmatrix} -1 & -1 & 10 \\ 0 & -1 & -1 \\ 0 & 0 & -1 \end{bmatrix}$
- ☐ (d) $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 10 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Risposta corretta.

The correct answer is: $\mathbf{A}^{-1} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}$

Question 5

Incorrect

Mark -15.00 out of 100.00

Let $f : \mathbb{R}^5 \rightarrow \mathbb{R}^5$ be an endomorphism with characteristic polynomial $P(t) = -(t^2 - 3)(t^3 - t)$.

Which one of the following statements is true?

- ☒ (a) $\dim(\text{Im}(f)) = 5$. ✗
- ☐ (b) $\dim(\text{Im}(f)) = 2$.
- ☐ (c) $\dim(\text{Ker}(f)) = 1$.
- ☐ (d) $\dim(\text{Ker}(f)) = 3$.

Risposta errata.

The correct answer is: $\dim(\text{Ker}(f)) = 1$.

Question 6

Incorrect

Mark -15.00 out of
100.00

In $\mathbb{R}^4$ let us consider the vector subspace

$$V = \{ (x, y, z, w) \mid x + y - z - w = 0, x - 2y = 0 \}.$$

Which one of the following statements is true?

- ☒ (a) V is a subspace of dimension 1. ✗
- ☐ (b) V is the vector subspace generated by the vectors $(1, 1, 1, 1)$ e $(0, 0, 1, 1)$.
- ☐ (c) V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$.
- ☐ (d) A basis of V is formed by 3 vectors.

Risposta errata.

The correct answer is: V is the vector subspace generated by the vectors $(2, 1, 2, 1)$ e $(0, 0, 1, -1)$.

Question 7

Incorrect

Mark -15.00 out of
100.00

Let us consider the matrix $A = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 1 & h \\ 0 & h & 4 \end{pmatrix}$

with $h \in \mathbb{R}$.

Which one of the following statements is true?

- ☒ (a) A is diagonalizable only if $h \neq 4$. ✗
- ☐ (b) A is diagonalizable only if $h = 0$.
- ☐ (c) A can be diagonalized for all $h \in \mathbb{R}$.
- ☐ (d) A has 3 distinct eigenvalues for all $h \in \mathbb{R}$.

Risposta errata.

The correct answer is: A can be diagonalized for all $h \in \mathbb{R}$.

Question 8

Incorrect

Mark -15.00 out of
100.00

In the euclidean 2-dimensional space with a fixed cartesian system, let h be a real parameter and consider the conic C_h having equation

$$4hx^2 + hy^2 + 4(h+1)xy + 2x = 0.$$

Which of the following statements is true?

-
- ☒ (a) C_h is a parabola for every $h \in \mathbb{R}$. ✗
- ☐ (b) C_h is an ellipse for every $h \in \mathbb{R}$.
- ☐ (c) C_h is degenerate for $h = 0$.
- ☐ (d) C_h is a hyperbola for every $h \in \mathbb{R}$.

Risposta errata.

The correct answer is: C_h is degenerate for $h = 0$.

Question 9

Incorrect

Mark -15.00 out of
100.00

In the euclidean 3-dimensional space with a fixed cartesian system, let us consider the sphere $\mathcal{S}$ and the plane π of equations respectively

$$\mathcal{S} : x^2 + y^2 + z^2 - 4z = 0, \quad \pi : x - z + 2 = 0.$$

Which of the following statements is true?

-
- ☒ (a) $(0, 2, 0) \in \mathcal{S} \cap \pi$. ✗
- ☐ (b) The distance of the center of $\mathcal{S}$ from the plane π is equal to 1.
- ☐ (c) The radius of the circumference $\mathcal{S} \cap \pi$ is equal to 2.
- ☐ (d) π is tangent to $\mathcal{S}$.

Risposta errata.

The correct answer is: The radius of the circumference $\mathcal{S} \cap \pi$ is equal to 2.

Question 10

Incorrect

Mark -15.00 out of
100.00

Let us consider the quadratic form defined by

$$f(x, y) = (x, y)A \begin{pmatrix} x \\ y \end{pmatrix} \quad \text{with} \quad A = \begin{pmatrix} 3 & 1 \\ 1 & 2 \end{pmatrix}.$$

Which of the following statements is true?

-
- ☒ (a) $q(x, y) - 1 = 0$ is the equation of a hyperbola. ✖
- ☐ (b) There exists a pair $(a, b) \in \mathbb{R}^2 \setminus \{(0, 0)\}$ such that $q(a, b) = 0$.
- ☐ (c) $q(x, y)$ is positive definite.
- ☐ (d) The matrix A has two eigenvalues of different sign.

Risposta errata.

The correct answer is: $q(x, y)$ is positive definite.